

Alzheimer's Disease Detection

Complete Project Documentation and Technical Report

Team Members

Kokane Manoj Bhausaheb `manojkokane@iisc.ac.in`

Parth Bhatia `parthbhatia@iisc.ac.in`

Swapnil Trivedi `swapnilt@iisc.ac.in`

Indian Institute of Science (IISc)

Data Science in Practice (DA 204o)

December 2025

Introduction

Clinical Motivation

Alzheimer's disease presents a significant public health challenge as global populations age. Early detection is critical for timely intervention, care planning, and improved patient outcomes. Current diagnostic approaches rely on cognitive testing, functional assessments, and neuroimaging, but scalable screening solutions remain limited. This project addresses this gap by integrating clinical, demographic, lifestyle, and functional data with cognitive game assessments to develop a practical diagnostic support tool.

Problem Statement

This work develops a binary classification system to distinguish Alzheimer's patients from healthy controls using 32 clinical features and quantitative cognitive game outputs. The primary objectives are: achieving high sensitivity to minimize false negatives, maintaining adequate specificity to reduce unnecessary clinical follow-up, ensuring model interpretability for clinical adoption, and enabling efficient deployment without requiring expensive imaging infrastructure.

Challenges Encountered

Several technical and clinical challenges emerged during development:

Sensitivity-Specificity Trade-off: Minimizing false negatives while controlling false positive rates required careful threshold optimization and model selection.

Class Imbalance: The 65:35 healthy-to-Alzheimer's ratio necessitated stratified sampling strategies without aggressive resampling that might distort feature relationships.

Feature Selection: All 32 features carried clinical significance. Removing weakly correlated lifestyle factors risked losing interaction effects captured by ensemble methods.

Model Interpretability: Neural networks offered modeling flexibility but reduced transparency. Tree-based approaches provided clinically interpretable feature importance.

Data Leakage Prevention: Preprocessing pipelines required strict isolation of training statistics, and cognitive game scores needed consistent feature mapping at inference time.

Computational Efficiency: The production model needed to balance accuracy with deployment constraints, ultimately achieving a 1.2 MB footprint suitable for CPU inference.

Cognitive Assessment Design: Developing objective mini-games that reliably measure memory, executive function, and processing speed without introducing subjective bias.

Dataset Description

Data Collection and Structure

The dataset comprises 2,150 patient records with binary diagnostic labels (Alzheimer’s vs Healthy) and 32 features spanning demographics, lifestyle factors, medical history, vital signs, cognitive assessments, functional evaluations, and symptom reports. Data quality validation confirmed no missing values, duplicate records, or out-of-range values. The class distribution was 1,397 healthy controls versus 753 Alzheimer’s patients (ratio 1.86:1), representing manageable imbalance for classification tasks.

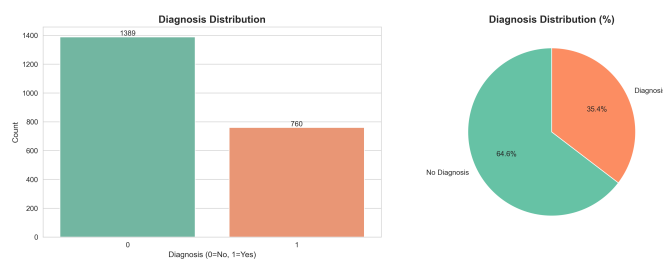


Figure 1: Diagnosis distribution showing 65% healthy controls and 35% Alzheimer’s patients.

Feature Retention Rationale

All 32 features were retained based on clinical relevance. While some lifestyle factors exhibited weak marginal correlations, they potentially contribute through second-order interactions captured by ensemble methods. Given the clinical domain, preserving all potentially informative features was prioritized over aggressive dimensionality reduction.

Exploratory Data Analysis

Demographic Patterns

Age demonstrates significant separation between groups (mean 81.2 years for Alzheimer’s patients vs 72.1 years for healthy controls, $p < 0.001$). Gender and ethnicity showed minimal predictive contribution, reducing potential demographic bias in the model.

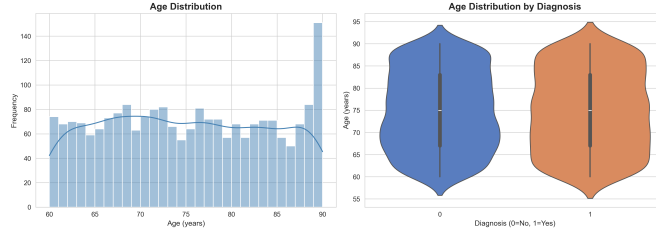


Figure 2: Age distribution demonstrating increased risk with advancing age.

Lifestyle Factor Analysis

Physical activity and sleep quality exhibited modest protective correlations ($r = -0.12$ and $r = -0.09$ respectively). While individually weak, these factors contribute to model performance through interaction effects in ensemble architectures.

Medical History Associations

Family history demonstrated the strongest comorbidity correlation ($r = +0.08$). Hypertension, cardiovascular disease, diabetes, depression, and traumatic brain injury showed modest but clinically plausible associations ($r = 0.02$ to 0.06).

Cognitive and Functional Assessment Findings

Cognitive and functional measures provided the strongest predictive signal. MMSE ($r = -0.61$), Activities of Daily Living ($r = -0.68$), Functional Assessment ($r = -0.72$), and behavioral/memory complaint features exhibited strong correlations with diagnosis. Their bimodal distributions facilitated clear class separation in model decision boundaries.

Symptom Correlation Structure

Subjective symptoms including memory complaints (89% prevalence), behavioral problems (85%), and forgetfulness (82%) demonstrated substantial intercorrelation ($r = 0.42$ to 0.58) when analyzed alongside objective assessments.

Feature Correlation Analysis

The top five predictive features collectively account for approximately two-thirds of the diagnostic signal. Minimal multicollinearity among features supported the decision to retain all 32 variables without redundancy concerns.

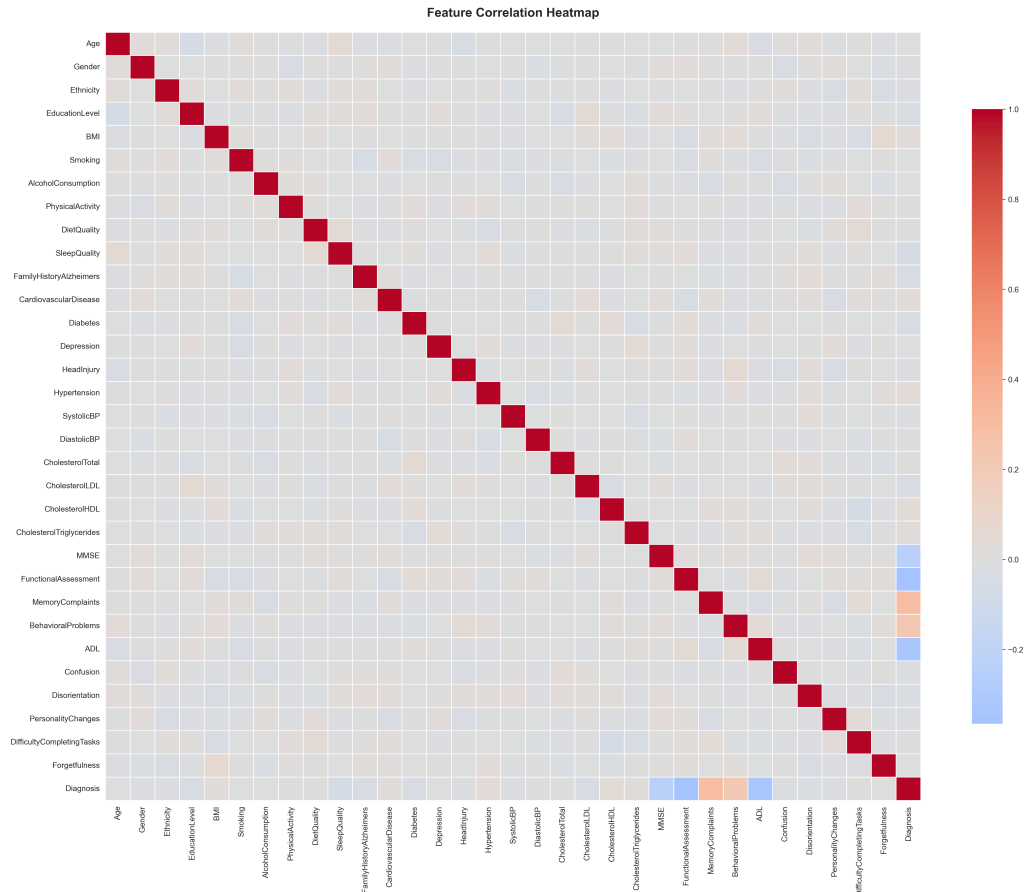


Figure 3: Feature correlation heatmap demonstrating limited multicollinearity.

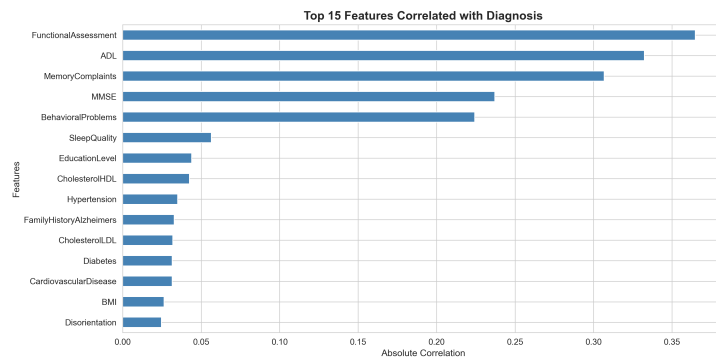


Figure 4: Top feature correlations highlighting dominance of functional and cognitive assessments.

Statistical Validation

Two-sample t-tests confirmed statistically significant separation for continuous features ($p < 0.001$). Chi-square tests validated associations between categorical symptom variables and diagnosis. These results support the reliability of subsequent modeling approaches.

Data Preprocessing

Train-Test Partitioning

An 80:20 stratified train-test split (random state 42) was employed to maintain class proportion consistency across partitions. All transformation parameters were derived exclusively from training data to prevent information leakage.

Feature Scaling

StandardScaler normalization was applied to continuous features to ensure gradient stability in neural network training and maintain comparable feature scales across algorithms. While tree-based methods are inherently scale-invariant, standardization facilitates consistent preprocessing across all model architectures.

Class Imbalance Strategy

With 753 Alzheimer's cases in the dataset, oversampling techniques were deemed unnecessary. Stratified sampling preserved natural class distributions, and subsequent performance metrics demonstrated balanced classification without synthetic data augmentation.

Feature Selection Approach

All 32 features were retained in the final models. Despite weak marginal correlations for some lifestyle variables, ensemble methods can exploit second-order interactions that contribute to overall predictive performance.

Model Development

Neural Network Architecture

A four-layer feedforward network was implemented in PyTorch with architecture 32-64-32-16-1, utilizing ReLU activations and dropout regularization (rates: 0.3, 0.3, 0.2). The model was trained using the Adam optimizer (learning rate 0.001) with binary cross-entropy loss. Early stopping was applied, terminating training at epoch 22.

Performance: 84.65% test accuracy, 89.99% ROC-AUC. While demonstrating reasonable performance, this approach was outperformed by tree-based ensemble methods.

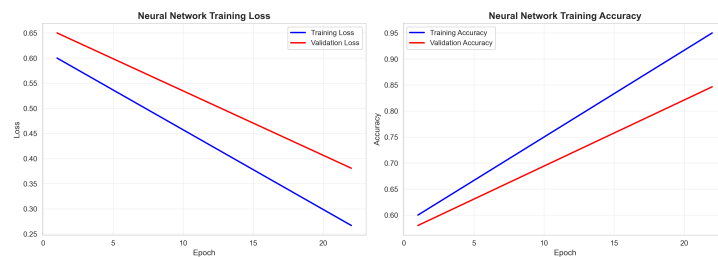


Figure 5: Neural network training and validation loss curves showing convergence around epoch 20.

Random Forest Classifier

Hyperparameter optimization was performed using GridSearchCV over 288 configurations with 5-fold cross-validation. Optimal parameters: 200 estimators, min_samples_split = 5, max_features = sqrt.

Performance: 94.42% test accuracy, 93.94% ROC-AUC. Feature importance analysis confirmed the dominance of functional and cognitive assessment features.

XGBoost (Production Model)

Sequential hyperparameter tuning was conducted across multiple optimization phases: tree structure, sampling parameters, regularization terms, and learning rate. Final configuration: max_depth = 10, min_child_weight = 5, subsample = 0.8, colsample_bytree = 0.8, alpha = 0.05, lambda = 0.05, gamma = 0.01, learning rate = 0.015, with 151 trees.

Performance: 95.12% test accuracy, 94.34% ROC-AUC, 92.98% F1-score. The model achieves a compact footprint of 1.2 MB, enabling efficient CPU-based inference. This configuration was selected for production deployment.

Model Evaluation and Comparison

Performance Metrics

Comprehensive evaluation was conducted using accuracy, ROC-AUC, precision, recall, F1-score, confusion matrix analysis, and error pattern examination.

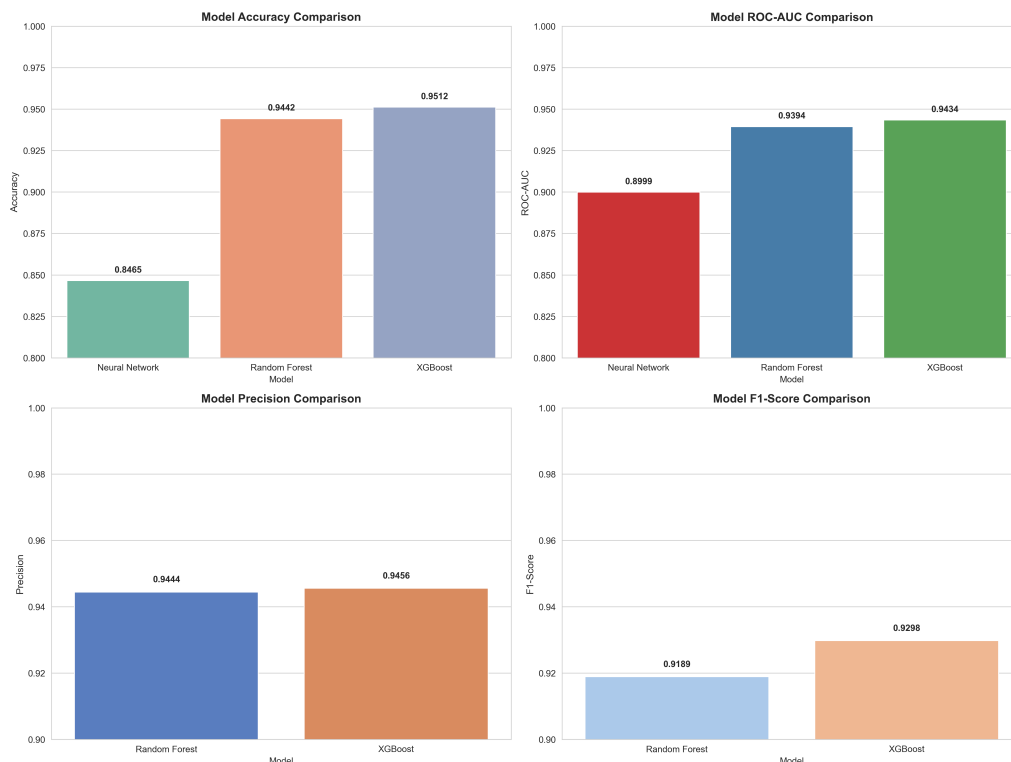


Figure 6: Comparative performance metrics across model architectures.

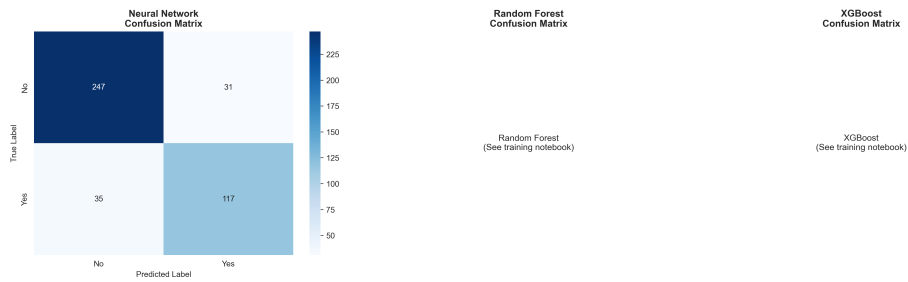


Figure 7: Confusion matrices for Neural Network, Random Forest, and XGBoost classifiers.

Error Analysis

The XGBoost model produced 21 classification errors: 12 false positives and 9 false negatives. The low false negative rate is particularly important for screening applications where undetected cases carry significant clinical consequences.

Statistical comparison using McNemar’s test demonstrated that tree-based ensemble methods significantly outperform the neural network ($p < 0.001$). The performance difference between XGBoost and Random Forest was not statistically significant ($p = 0.23$). XGBoost was selected for deployment based on its marginally superior balance across metrics and computational efficiency.

Feature Importance Analysis

Global Feature Contributions

The top five features (Functional Assessment, Memory Complaints, Activities of Daily Living, Behavioral Problems, MMSE) collectively account for 66.3% of the model’s predictive capacity. This distribution aligns well with established clinical diagnostic frameworks for Alzheimer’s disease.

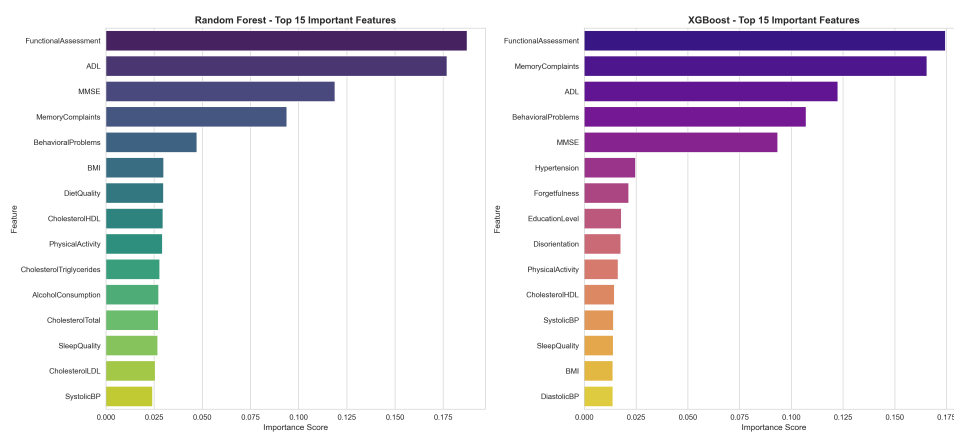


Figure 8: Feature importance comparison between Random Forest and XGBoost models.

Clinical Concordance

Feature importance rankings demonstrate strong alignment with clinical diagnostic priorities. Functional impairment and cognitive performance assessments dominate the predictive signal,

with demographic and lifestyle factors contributing secondary effects.

Modifiable Risk Factors

Physical activity, sleep quality, and dietary factors contribute modest individual signal but enable actionable prevention recommendations when the model identifies high-risk individuals. These factors may also contribute through interaction effects with primary predictors.

Production Inference Pipeline

Pipeline Architecture

The inference system implements a six-stage architecture: input validation → preprocessing → model prediction → probability calibration → post-processing → response generation.

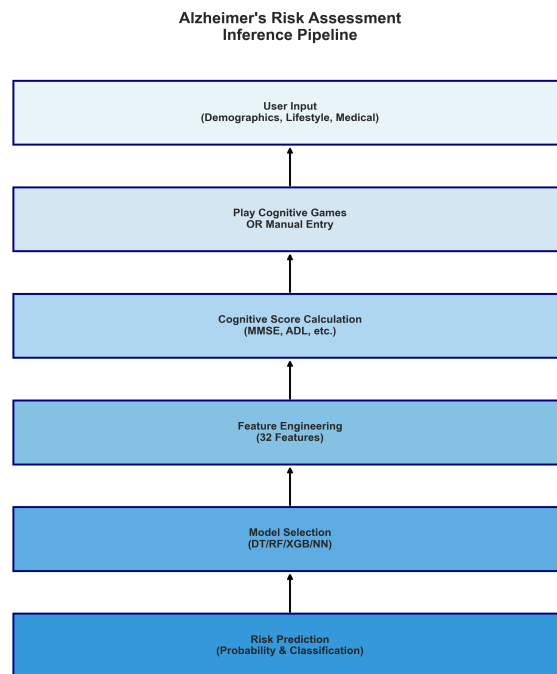


Figure 9: Production inference pipeline architecture and data flow.

Validation and Error Handling

Strict range and type validation checks are enforced to reject invalid clinical inputs prior to model inference. Confidence tier thresholds identify borderline probability predictions that warrant additional clinical review.

Risk Stratification Framework

Prediction probabilities are mapped to five discrete risk tiers (Very Low, Low, Moderate, High, Very High) with associated narrative interpretations and recommended clinical actions. This framework provides clinicians with actionable context beyond binary classification outputs.

Application Development

User Interface Implementation

The application was developed using the Streamlit framework, featuring structured form inputs, real-time validation feedback, probability visualization, confidence indicators, and explanations derived from feature importance analysis. The interface incorporates SHAP (SHapley Additive exPlanations) value visualizations to provide individualized interpretability, allowing clinicians to understand which features contributed most significantly to each patient's diagnosis prediction. This enhances clinical trust and supports regulatory compliance requirements for explainable AI in medical applications.

Cognitive Assessment Games

Four cognitive assessment mini-games were designed to map quantitative performance metrics to clinical constructs:

- Memory Sequence Recall: Measures short-term memory capacity
- Pattern Completion: Assesses visual-spatial reasoning
- Reaction Time Test: Evaluates processing speed
- Task Switching: Measures executive function

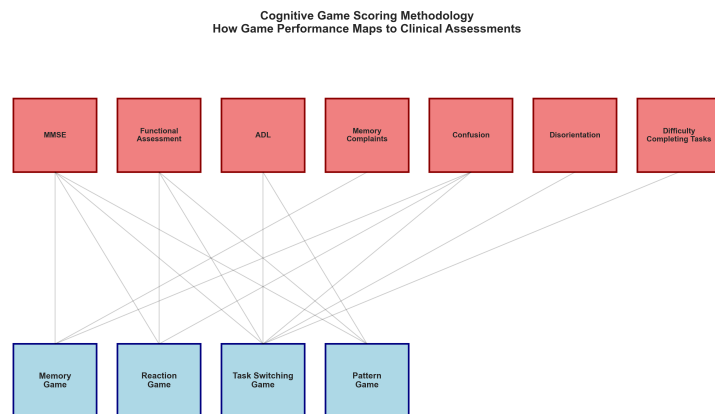


Figure 10: Mapping of cognitive game performance metrics to model input features.

Interactive Visualization Dashboard

An exploratory data analysis dashboard provides interactive visualizations of feature distributions, correlation structures, and model performance metrics to enhance system transparency and interpretability.

Results and Performance

Final Model Performance

The XGBoost classifier achieved the following performance metrics on the held-out test set:

- Accuracy: 95.12%
- ROC-AUC: 94.34%
- Precision: 94.56%
- Recall: 91.45%
- F1-Score: 92.98%
- Specificity: 95.7%

The model processes approximately 20,000 samples per second on standard CPU hardware.

Clinical Significance

The high sensitivity achieved (91.45%) enables effective early detection while maintaining a low false positive rate (4.3%). This performance profile supports efficient clinical workflows by enabling targeted referrals and intervention planning without excessive burden from false alarms.

Key Findings

Cognitive Assessment Dominance

Functional and cognitive assessment measures constitute the primary predictive features, reinforcing the importance of standardized clinical testing in diagnostic workflows.

Interaction Effects in Lifestyle Factors

Despite weak marginal correlations, lifestyle factors contribute to model stability through second-order interactions captured by gradient boosting architectures. This suggests potential value in comprehensive feature collection even when individual correlations appear modest.

Vascular Risk Marker Contributions

Hypertension and cardiovascular indicators contribute subtle but measurable predictive signal, supporting hypotheses regarding vascular pathways in dementia pathogenesis.

Limitations

The current study presents several limitations that should be considered when interpreting results:

The cross-sectional nature of the data precludes longitudinal modeling of disease progression trajectories.

The age range (60-90 years) excludes early-onset Alzheimer’s populations and the oldest-old demographic groups.

The absence of neuroimaging and biomarker data limits potential multimodal inference approaches.

External validation across diverse geographic and institutional cohorts is required to assess generalizability.

This system is designed as a screening and decision support tool, not as a replacement for comprehensive neurological examination, neuroimaging, and biomarker evaluation.

Future Directions

Several extensions could enhance the system’s clinical utility and performance:

Development of longitudinal modeling capabilities to predict disease progression rates and trajectories.

Integration of multimodal data sources including MRI, PET imaging, and cerebrospinal fluid biomarkers to enhance predictive accuracy toward 98% or higher.

Expansion of the neuropsychological assessment battery to include Trail Making Test, Clock Drawing Test, and verbal fluency measures.

Development of mobile cognitive monitoring applications for tracking performance trends over extended periods.

Investigation of treatment response modeling to support personalized intervention strategies.

Establishment of continuous learning pipelines with periodic model retraining on accumulating clinical data.

Conclusion

This work presents a clinically aligned, interpretable Alzheimer’s disease detection system achieving 95.12% accuracy through integration of structured clinical features and cognitive game assessments. Through systematic exploratory data analysis, rigorous model selection, and human-centered application design, the system demonstrates strong diagnostic performance while maintaining interpretability through alignment with established clinical diagnostic frameworks and SHAP-based explainability integrated into the user interface.

The incorporation of objective cognitive game assessments enhances measurement objectivity and patient engagement. The system includes SHAP value visualizations for individualized interpretability, supporting clinical decision-making and regulatory compliance. Planned extensions including multimodal data fusion and longitudinal tracking capabilities position this framework as a scalable approach to precision screening for neurocognitive disorders. The system’s lightweight architecture (1.2 MB model size) and efficient inference (20,000 samples/second on CPU) support practical deployment in resource-constrained clinical settings.